Typography-Aware Text-to-Image Generation: Toward Controllable Poster and Text-Heavy Visual Synthesis
6.S058 Final Project — Draft Introduction

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1. Introduction

Modern text-to-image (T2I) diffusion models such as Stable Diffusion [13], DALL·E 3 [1], and Imagen [14] can synthesize photorealistic and artistically elaborate images from natural-language prompts. Yet almost every practitioner who has tried to generate a movie poster, a book cover, a social-media banner, or an advertisement has run into the same wall: these systems produce beautiful *imagery*, but when the visual must *contain text*, the results are unusable. Titles are misspelled or illegible, captions are placed seemingly at random, type weight and color clash with the background, and the overall composition ignores basic rules of graphic design. The failure is not incidental to scale—it persists in 2026-era models—because text inside an image is simultaneously a semantic, a typographic, and a spatial object, and current T2I models have no inductive bias that treats it as such.

We focus on this gap: *controllable, typography-aware image generation for text-heavy visuals*. We view it as both a computer-vision problem and a generative-modeling problem. On the CV side, it sits next to long-standing work on scene text [12] and document understanding [8], but in reverse: instead of *reading* text in images, the model must *place and render* it legibly and aesthetically. On the generative side, it forces us to move beyond the single prompt $\rightarrow$ single image pipeline toward *structured, layered* generation in which typography is a first-class citizen.

Value to the field. Text-in-image synthesis is the bottleneck for an entire class of real-world design applications—marketing assets, e-commerce thumbnails, slide decks, zines, educational infographics—most of which are currently handled either by professional designers or by template-based tools such as Canva. Closing this gap would move T2I from illustration toward genuine creative production, and it would push the community to build richer, more compositional image representations. For computer-vision research specifically, the task demands grounding language into spatial, hierarchical, editable visual structure; progress here is likely to transfer to layout understanding, document-image synthesis for OCR training, and graphical user interface generation.

Related work and the gap we target. Three lines of work are closest. First, **text-rendering methods** such as GlyphControl [16], TextDiffuser [2], AnyText [15], and Glyph-ByT5 [11] inject explicit glyph or character-level signals into the diffusion model and have substantially improved spelling inside images; several of them, notably TextDiffuser and AnyText, additionally support forms of text inpainting and editing. However, although these methods substantially improve text fidelity, they typically do not expose typography as a persistent structured layer with full downstream editability and global design-level control: font family, weight, tracking, and the relationship between text and underlying imagery are not first-class, re-configurable attributes of the output. Second, **layout-aware generation** methods such as GLIGEN [9], ControlNet [17], and LayoutGPT [3] condition diffusion on bounding boxes and semantic tags, but they operate at the object level rather than at the typographic level—font, weight, tracking, and type hierarchy are simply not part of the interface. Third, **poster and design-layout methods**, including PosterLayout [6], CGL-GAN [19], and AutoPoster [10], decouple layout from rendering: they first predict element boxes and then rasterize the visual with a separate, often lower-quality generator. None of these three lines jointly reasons about *imagery*, *typography as a structured layer*, and *aesthetic composition*, and evaluation has continued to rely on FID [5] and CLIPScore [4], neither of which captures text fidelity or design controllability.

Our approach. We build a lightweight control adapter [7, 17] on top of a pretrained text-aware T2I backbone such

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as TextDiffuser [2] or AnyText [15], exposing a structured edit interface: for a single text element in the image, the user specifies its content, bounding box, font size, and color, and the adapter regenerates only the requested change while keeping the surrounding image stable. We focus on these four text attributes because they are the most frequent targets of design-time edits and admit clean quantitative measurement; extending the interface to font family, weight, tracking, and full typographic hierarchy is a natural follow-up. For supervision we use synthetic poster renderings—backgrounds drawn from permissively licensed image collections composited with programmatically rendered text layouts—so that each training example comes with exact attribute labels and a ground-truth pre-/post-edit pair, without requiring typography recovery from real designs. We evaluate along four axes using existing metrics: OCR accuracy via a frozen recognizer for text fidelity, an edit success rate that measures whether the recognized text at the requested box matches the requested content, LPIPS [18] and SSIM between the pre- and post-edit images outside the text box for background preservation, and a small human A/B preference study. Baselines include the unmodified backbone, a prompt-only regeneration baseline, and a ControlNet-style layout-only conditioning baseline [17].

2. Use of Generative AI

I used Generative AI tools (Claude Opus 4.7) to manage the citations and the CVPR formatting.

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